Exercise Set 1.4 Solution

Question 11

Find the inverse of

$$\begin{bmatrix} \cos\theta & \sin\theta \\ -\sin\theta & \cos\theta \end{bmatrix}$$

.

Solution: The matrix $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$ is invertible if $ad - bc \neq 0$, in which case the inverse is given by the formula

$$A^{-1} = \frac{1}{ad-bc}\begin{bmatrix} d & -b \\ -c & a \end{bmatrix} = \begin{bmatrix} \frac{d}{ad-bc} & -\frac{b}{ad-bc} \\ -\frac{c}{ad-bc} & \frac{a}{ad-bc} \end{bmatrix}$$

In our case $A = \begin{bmatrix} \cos\theta & \sin\theta \\ -\sin\theta & \cos\theta \end{bmatrix}$

$$A^{-1} = \frac{1}{\cos\theta^2 + \sin\theta^2}\begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix} = \begin{bmatrix} \frac{\cos\theta}{\cos\theta^2+\sin\theta^2} & -\frac{\sin\theta}{\cos\theta^2+\sin\theta^2} \\ \frac{\sin\theta}{\cos\theta^2+\sin\theta^2} & \frac{\cos\theta}{\cos\theta^2+\sin\theta^2} \end{bmatrix}$$

$$A^{-1} = \begin{bmatrix} \frac{\cos\theta}{1} & -\frac{\sin\theta}{1} \\ \frac{\sin\theta}{1} & \frac{\cos\theta}{1} \end{bmatrix} = \begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix}$$

Question 13

Consider the matrix

$$A = \begin{bmatrix} a_{11} & 0 & \cdots & 0 \\ 0 & a_{22} & \cdots & 0 \\ \vdots & \vdots & & \vdots \\ 0 & 0 & \cdots & a_{nn} \end{bmatrix}$$

where $a_{11}, a_{22}, \ldots, a_{nn} \neq 0$. Show that A is invertible and find its inverse.

Solution: Definition of Inverse $A^{-1}A = AA^{-1} = I$

If we operate A by matrix given below

$$\begin{bmatrix} \frac{1}{a_{11}} & 0 & 0 & \cdots & 0 \\ 0 & \frac{1}{a_{22}} & 0 & \cdots & 0 \\ 0 & 0 & \frac{1}{a_{33}} & \cdots & 0 \\ \vdots & \vdots & \vdots & & \vdots \\ 0 & 0 & 0 & \cdots & \frac{1}{a_{nn}} \end{bmatrix}$$

We will have identity matrix.

$$\begin{bmatrix} \frac{1}{a_{11}} & 0 & 0 & \cdots & 0 \\ 0 & \frac{1}{a_{22}} & 0 & \cdots & 0 \\ 0 & 0 & \frac{1}{a_{33}} & \cdots & 0 \\ \vdots & \vdots & \vdots & & \vdots \\ 0 & 0 & 0 & \cdots & \frac{1}{a_{nn}} \end{bmatrix} \begin{bmatrix} a_{11} & 0 & \cdots & 0 \\ 0 & a_{22} & \cdots & 0 \\ \vdots & \vdots & & \vdots \\ 0 & 0 & \cdots & a_{nn} \end{bmatrix} = \begin{bmatrix} 1 & 0 & \cdots & 0 \\ 0 & 1 & \cdots & 0 \\ \vdots & \vdots & & \vdots \\ 0 & 0 & \cdots & 1 \end{bmatrix}$$

$$\begin{bmatrix} a_{11} & 0 & \cdots & 0 \\ 0 & a_{22} & \cdots & 0 \\ \vdots & \vdots & & \vdots \\ 0 & 0 & \cdots & a_{nn} \end{bmatrix} \begin{bmatrix} \frac{1}{a_{11}} & 0 & 0 & \cdots & 0 \\ 0 & \frac{1}{a_{22}} & 0 & \cdots & 0 \\ 0 & 0 & \frac{1}{a_{33}} & \cdots & 0 \\ \vdots & \vdots & \vdots & & \vdots \\ 0 & 0 & 0 & \cdots & \frac{1}{a_{nn}} \end{bmatrix} = \begin{bmatrix} 1 & 0 & \cdots & 0 \\ 0 & 1 & \cdots & 0 \\ \vdots & \vdots & & \vdots \\ 0 & 0 & \cdots & 1 \end{bmatrix}$$

Question 15

(a) Show that a matrix with a row of zeros cannot have an inverse.

(b) Show that a matrix with a column of zeros cannot have an inverse.

Solution for both part: Let A denote a matrix which has an entire row or an entire column of zeros. Then if B is any matrix, either AB has an entire row of zeros or BA has an entire column of zeros, respectively. (See Exercise 18, Section 1.3.) Hence, neither AB nor BA can be the identity matrix; therefore, A cannot have an inverse.

Question 29

(a) Show that if A is invertible and $AB = AC$, then $B = C$.

Solution: Given $AB = AC$ and A^{-1} exists. Apply A^{-1} from L.H.S.

$$\begin{aligned} A^{-1}AC &= A^{-1}C \\ IB &= IC \qquad \qquad \therefore \quad I \text{ is an identity matrix.} \\ B &= C. \end{aligned}$$

(b) Explain why part (a) and Example 3 do not contradict one another.

Solution: $A = \begin{bmatrix} 0 & 1 \\ 0 & 2 \end{bmatrix}$ and A is not invertible, i.e. A^{-1} does not exist.
One can see Det $A = 0.2 - 1.0 = 0$.

$$A^{-1} = \frac{1}{0} \begin{bmatrix} 2 & -1 \\ 0 & 0 \end{bmatrix}$$

It is undefined.

Exercise Set 1.5 Solution

Question 02

Find a row operation that will restore the given elementary matrix to an identity matrix.

(a) $\begin{bmatrix} 1 & 0 \\ -3 & 1 \end{bmatrix}$

Solution: Multiplying Row 1 by 3 and adding to the Row 2.

$$\begin{bmatrix} 1 & 0 \\ -3 & 1 \end{bmatrix} (R_2 + 3 * R_1) \to \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

(b) $\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 3 \end{bmatrix}$

Solution: Multiplying Row 3 by $\frac{1}{3}$.

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 3 \end{bmatrix} (\frac{1}{3} * R_3) \to \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

(c) $\begin{bmatrix} 0 & 0 & 0 & 1 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 1 & 0 & 0 & 0 \end{bmatrix}$

Solution: Row Interchange by $R_1 \leftrightarrows R_4$.

(d) $\begin{bmatrix} 1 & 0 & -\frac{1}{7} & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$

Solution: Multiplying Row 3 by 7 and adding to the Row 1.

Question 03

Consider the matrices

$$A = \begin{bmatrix} 3 & 4 & 1 \\ 2 & -7 & -1 \\ 8 & 1 & 5 \end{bmatrix}, \quad B = \begin{bmatrix} 8 & 1 & 5 \\ 2 & -7 & -1 \\ 3 & 4 & 1 \end{bmatrix}, \quad C = \begin{bmatrix} 3 & 4 & 1 \\ 2 & -7 & -1 \\ 2 & -7 & 3 \end{bmatrix}$$

Find elementary matrices E_1, E_2, E_3, and E_4 such that

(a) $E_1 A = B$ Solution: If we interchange Rows 1 and 3 of A, then we obtain B. Therefore, E_1 must be the matrix obtained from I_3 by interchanging Rows 1 and 3 of I_3, i.e.,

$$E_1 = \begin{bmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{bmatrix}$$

(b) $E_2 B = A$ Solution: If we interchange Rows 1 and 3 of B, then we obtain A.

$$E_2 = \begin{bmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{bmatrix}$$

(c) $E_3A = C$

Solution: If we multiply Row 1 by -2 and add to Row 3 in A, then we obtain C.

$$E_3 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ -2 & 0 & 1 \end{bmatrix}$$

(d) $E_4C = A$

Solution: If we multiply Row 1 by 2 and add to Row 3 in C, then we obtain A.

$$E_4 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 2 & 0 & 1 \end{bmatrix}$$

Question 09

Find the inverse of each of the following 4×4 matrices, where k_1, k_2, k_3, k_4 and k are all nonzero.

(a) $\begin{bmatrix} k_1 & 0 & 0 & 0 \\ 0 & k_2 & 0 & 0 \\ 0 & 0 & k_3 & 0 \\ 0 & 0 & 0 & k_4 \end{bmatrix}$

Solution: Using $[A|I] \to [I|A^{-1}]$

$$\frac{1}{k_1} * R_1 \left[\begin{array}{cccc|cccc} 1 & 0 & 0 & 0 & \frac{1}{k_1} & 0 & 0 & 0 \\ 0 & k_2 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & k_3 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & k_4 & 0 & 0 & 0 & 1 \end{array}\right]$$

$$\frac{1}{k_2} * R_2 \left[\begin{array}{cccc|cccc} 1 & 0 & 0 & 0 & \frac{1}{k_1} & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & \frac{1}{k_2} & 0 & 0 \\ 0 & 0 & k_3 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & k_4 & 0 & 0 & 0 & 1 \end{array}\right]$$

$$\frac{1}{k_3} * R_3 \left[\begin{array}{cccc|cccc} 1 & 0 & 0 & 0 & \frac{1}{k_1} & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & \frac{1}{k_2} & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & \frac{1}{k_3} & 0 \\ 0 & 0 & 0 & k_4 & 0 & 0 & 0 & 1 \end{array}\right]$$

$$\frac{1}{k_4} * R_4 \left[\begin{array}{cccc|cccc} 1 & 0 & 0 & 0 & \frac{1}{k_1} & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & \frac{1}{k_2} & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & \frac{1}{k_3} & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & \frac{1}{k_4} \end{array}\right]$$

Hence $A^{-1} = \begin{bmatrix} \frac{1}{k_1} & 0 & 0 & 0 \\ 0 & \frac{1}{k_2} & 0 & 0 \\ 0 & 0 & \frac{1}{k_3} & 0 \\ 0 & 0 & 0 & \frac{1}{k_4} \end{bmatrix}$

(b) $\begin{bmatrix} 0 & 0 & 0 & k_1 \\ 0 & 0 & k_2 & 0 \\ 0 & k_3 & 0 & 0 \\ k_4 & 0 & 0 & 0 \end{bmatrix}$

Solution: Using $[A|I] \to [I|A^{-1}]$

$$R_1 \leftrightarrows R_4 \left[\begin{array}{cccc|cccc} k_4 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & k_2 & 0 & 0 & 1 & 0 & 0 \\ 0 & k_3 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & k_1 & 1 & 0 & 0 & 0 \end{array}\right]$$

$$R_3 \leftrightarrows R_3 \left[\begin{array}{cccc|cccc} k_4 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & k_3 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & k_2 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & k_1 & 1 & 0 & 0 & 0 \end{array}\right]$$

$$\frac{1}{k_4} * R_1 \left[\begin{array}{cccc|cccc} 1 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{1}{k_4} \\ 0 & k_3 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & k_2 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & k_1 & 1 & 0 & 0 & 0 \end{array}\right]$$

$$\frac{1}{k_3} * R_2 \left[\begin{array}{cccc|cccc} 1 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{1}{k_4} \\ 0 & 1 & 0 & 0 & 0 & 0 & \frac{1}{k_3} & 0 \\ 0 & 0 & k_2 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & k_1 & 1 & 0 & 0 & 0 \end{array}\right]$$

$$\frac{1}{k_2} * R_3 \left[\begin{array}{cccc|cccc} 1 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{1}{k_4} \\ 0 & 1 & 0 & 0 & 0 & 0 & \frac{1}{k_3} & 0 \\ 0 & 0 & 1 & 0 & 0 & \frac{1}{k_2} & 0 & 0 \\ 0 & 0 & 0 & k_1 & 1 & 0 & 0 & 0 \end{array}\right]$$

$$\frac{1}{k_1} * R_4 \left[\begin{array}{cccc|cccc} 1 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{1}{k_4} \\ 0 & 1 & 0 & 0 & 0 & 0 & \frac{1}{k_3} & 0 \\ 0 & 0 & 1 & 0 & 0 & \frac{1}{k_2} & 0 & 0 \\ 0 & 0 & 0 & 1 & \frac{1}{k_1} & 0 & 0 & 0 \end{array}\right]$$

Question 10

Consider the matrix

$$A = \begin{bmatrix} 1 & 0 \\ -5 & 2 \end{bmatrix}$$

(a) Find elementary matrices E_1 and E_2 such that $E_2 E_1 A = I$.
Solution: $E_2 E_1 A = I$

$$\begin{bmatrix} 1 & 0 \\ 0 & 1/2 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 5 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ -5 & 2 \end{bmatrix} = I_2$$

(b) Write A^{-1} as a product of two elementary matrices.
Solution: $E_2 E_1 A = (E_2 E_1) A = A^{-1} A = I$

$$\begin{bmatrix} 1 & 0 \\ 0 & \frac{1}{2} \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 5 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ \frac{5}{2} & \frac{1}{2} \end{bmatrix}$$

(c) Write A as a product of two elementary matrices.
Solution: $E_2 E_1 A = I \longrightarrow A = E_1^{-1} E_2^{-1} I = E_1^{-1} E_2^{-1}$

$$\begin{bmatrix} 1 & 0 \\ -5 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & 2 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ -5 & 2 \end{bmatrix}$$

Question 16

Show that

$$A = \begin{bmatrix} 0 & a & 0 & 0 & 0 \\ b & 0 & c & 0 & 0 \\ 0 & d & 0 & e & 0 \\ 0 & 0 & f & 0 & g \\ 0 & 0 & 0 & h & 0 \end{bmatrix}$$

is not invertible for any values of the entries.

Solution:

$$R_1 \leftrightarrows R_2 \left[\begin{array}{ccccc|ccccc} b & 0 & c & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & a & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & d & 0 & e & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & f & 0 & g & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & h & 0 & 0 & 0 & 0 & 0 & 1 \end{array}\right]$$

$$R_4 \leftrightarrows R_5 \left[\begin{array}{ccccc|ccccc} b & 0 & c & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & a & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & d & 0 & e & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & h & 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & f & 0 & g & 0 & 0 & 0 & 1 & 0 \end{array}\right]$$

$$R_3 - \frac{d}{a}R_2 \left[\begin{array}{ccccc|ccccc} b & 0 & c & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & a & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & e & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & h & 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & f & 0 & g & 0 & 0 & 0 & 1 & 0 \end{array}\right]$$

$$R_3 - \frac{e}{h}R_4 \left[\begin{array}{ccccc|ccccc} b & 0 & c & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & a & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & h & 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & f & 0 & g & 0 & 0 & 0 & 1 & 0 \end{array}\right]$$

One can see the third row is zero but the corresponding row of identity matrix is not zero. Hence it is inconsistent for any values of $a, b, c, d, e, f, g,$ and h.